deeplearning.ai

Recurrent Neural Networks

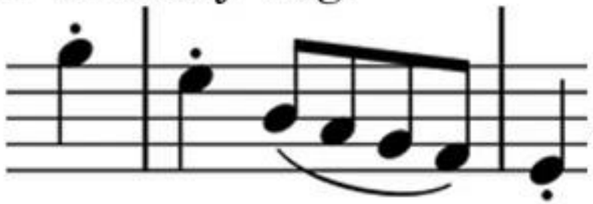
Why sequence models?

Examples of sequence data

x

y

Speech recognition → → "The quick brown fox jumped over the lazy dog."

Music generation → ∅ → 

Sentiment classification "There is nothing to like in this movie." → ★☆☆☆☆

DNA sequence analysis → AGCCCCTGTGAGGAACTAG → AGCCCCTGTGAGGAACTAG

Machine translation Voulez-vous chanter avec moi? → Do you want to sing with me?

Video activity recognition → Running

Name entity recognition → Yesterday, Harry Potter met Hermione Granger. → Yesterday, Harry Potter met Hermione Granger.

Andrew Ng

deeplearning.ai

Recurrent Neural Networks

Notation

Motivating example

NLP

x: Harry Potter and Hermione Granger invented a new spell.

$x^{<1>}$ $x^{<2>}$ $x^{<3>}$ .... $x^{<t>}$ .... $x^{<9>}$

$T_x = 9$

y: 1 1 0 1 1 0 0 0 0

$y^{<1>}$ $y^{<2>}$ $y^{<3>}$ .... $y^{<9>}$

$T_y = 9$

$x^{(i)<t>}$ $T_x^{(i)} = 9$ 15

$y^{(i)<t>}$ $T_y^{(i)}$

Andrew Ng

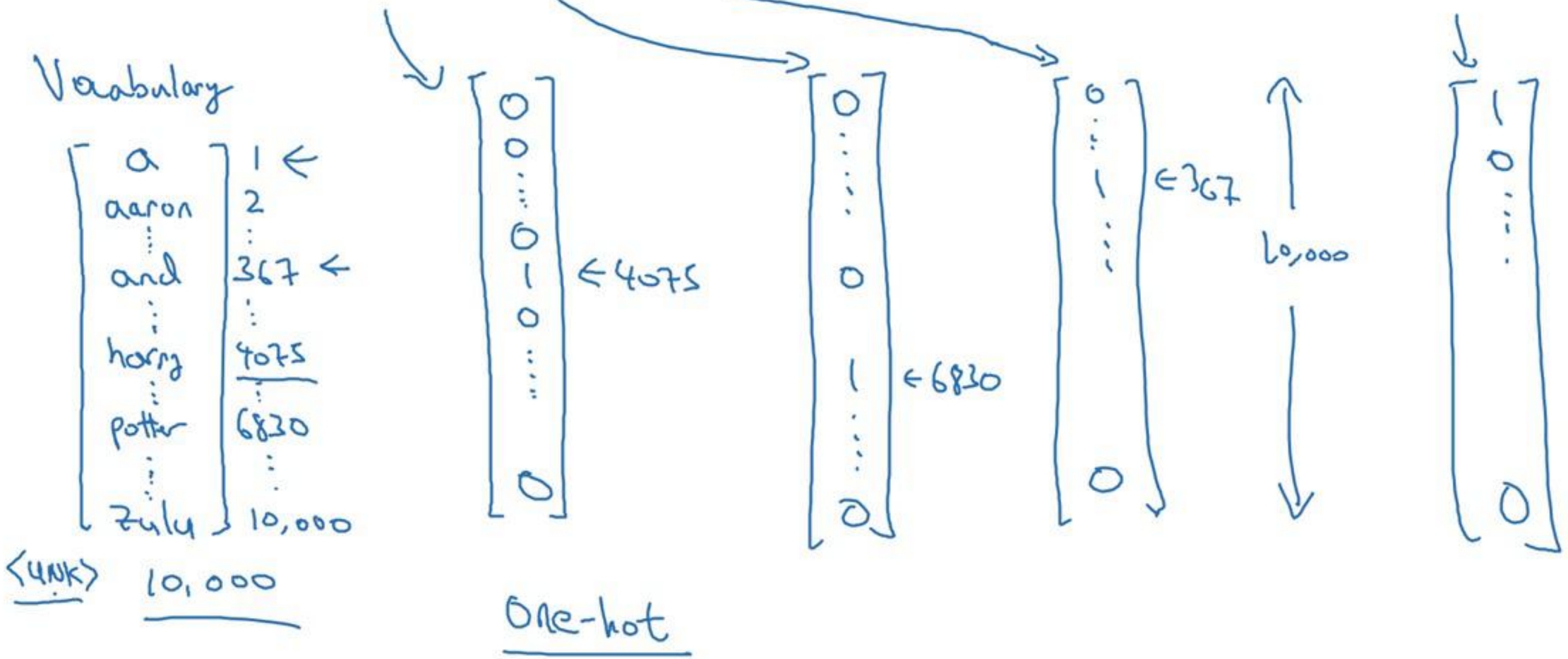
Representing words

$x^{<t>}$ (x, y)

$x \rightarrow y$

x: Harry Potter and Hermione Granger invented a new spell.

$x^{<1>}$ $x^{<2>}$ $x^{<3>}$ … $x^{<7>}$ $x^{<9>}$



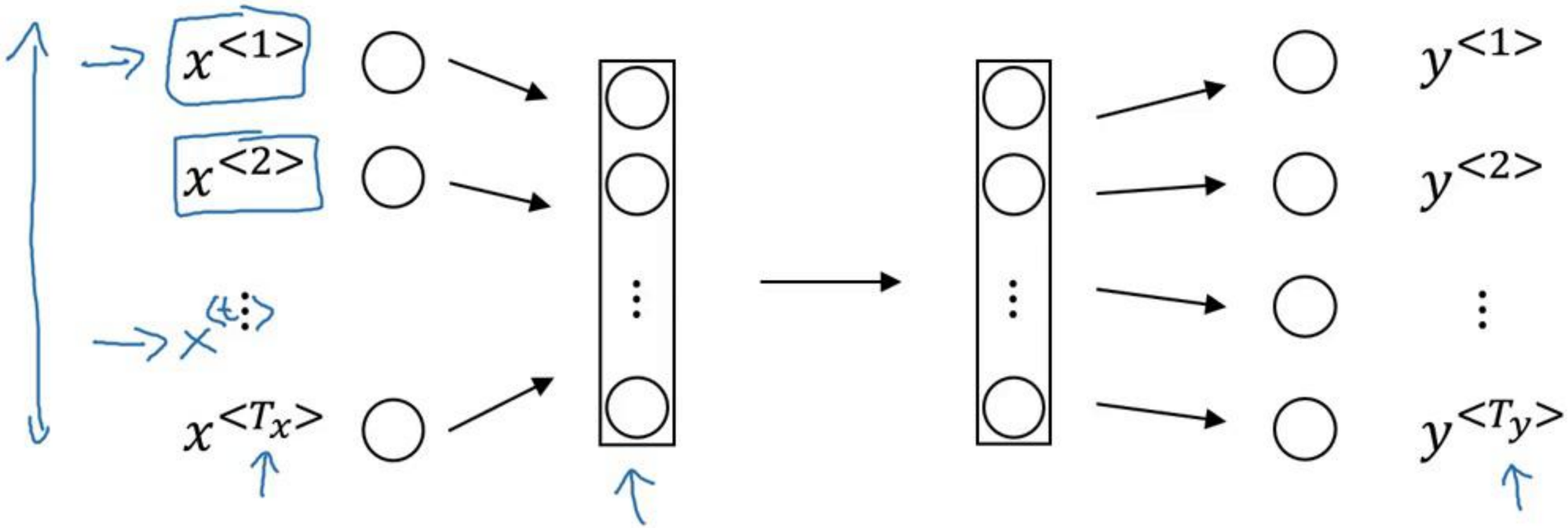
Andrew Ng

deeplearning.ai

Recurrent Neural Networks

Recurrent Neural Network Model

Why not a standard network?

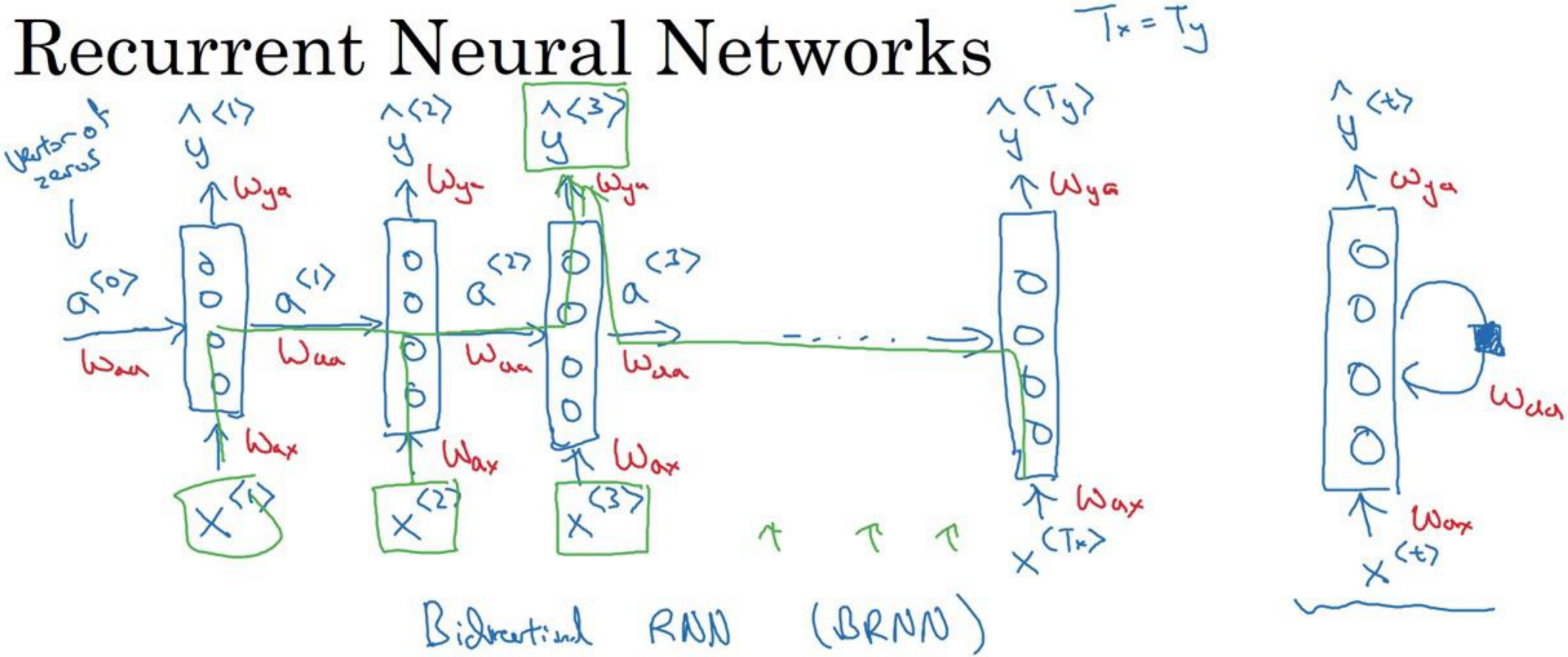


Problems:

- Inputs, outputs can be different lengths in different examples.
- Doesn't share features learned across different positions of text.

Andrew Ng

Recurrent Neural Networks



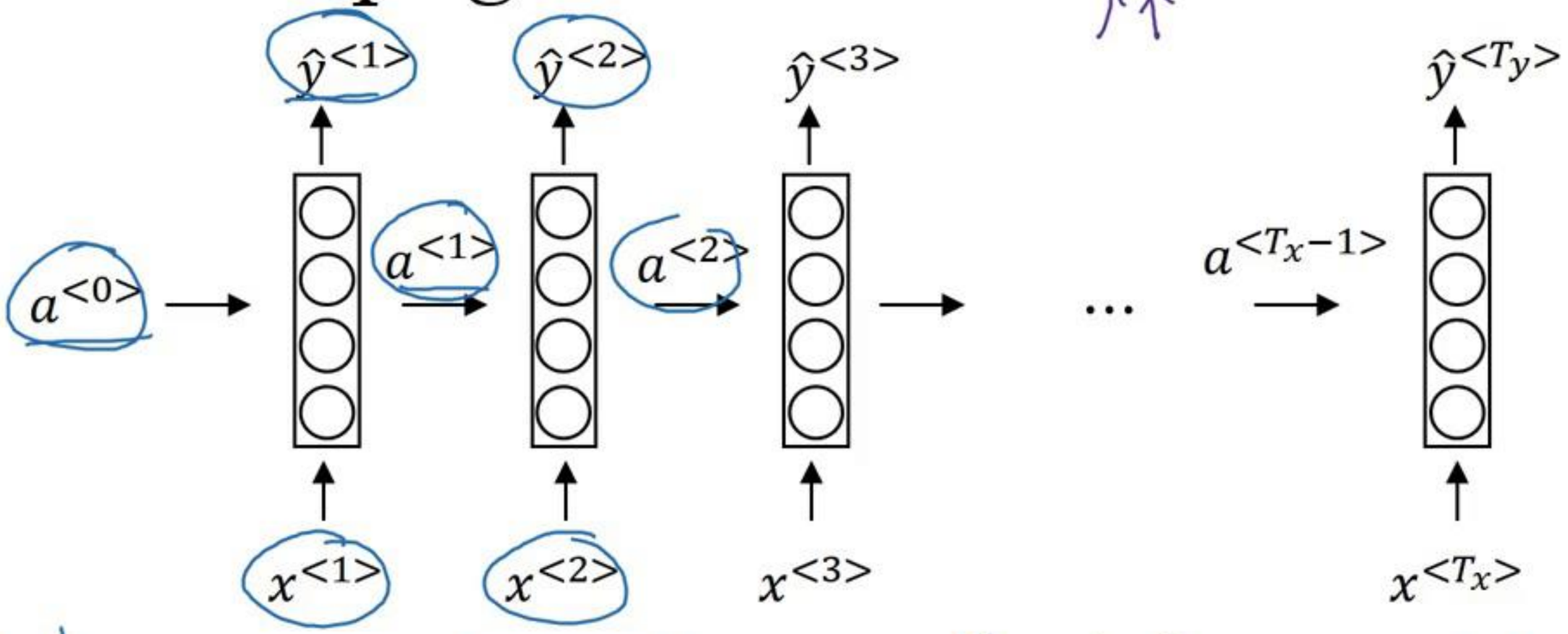
He said, "Teddy Roosevelt was a great President."

He said, "Teddy bears are on sale!"

Andrew Ng

Forward Propagation

$a \leftarrow W_{ax} x^{<1>}$



$a^{<0>} = \vec{0}$.

$a^{<1>} = g_1(W_{aa} a^{<0>} + W_{ax} x^{<1>} + b_a) \leftarrow$ tanh / ReLU

$\hat{y}^{<1>} = g_2(W_{ya} a^{<1>} + b_y) \leftarrow$ sigmoid

$$\boxed{\begin{aligned} a^{<t>} &= g(W_{aa} a^{<t-1>} + W_{ax} x^{<t>} + b_a) \\ \hat{y}^{<t>} &= g(W_{ya} a^{<t>} + b_y) \end{aligned}}$$

Andrew Ng

Simplified RNN notation

$$a^{<t>} = g(\boxed{W_{aa}}a^{<t-1>} + \boxed{W_{ax}}x^{<t>} + b_a)$$

100 10,000

(100,100) (100,10,000)

$$\hat{y}^{<t>} = g(W_{ya}a^{<t>} + b_y)$$

$$\hat{y}^{<t>} = g(W_y a^{<t>} + b_y)$$

$$a^{<t>} = g(W_a \boxed{[a^{<t-1>}, x^{<t>}]} + b_a)$$

$$\boxed{\begin{array}{c} 100 \updownarrow \\ \end{array}\begin{bmatrix} W_{aa} \,\vdots\, W_{ax} \end{bmatrix}} = W_a$$

100 10000

(100, 10100)

$$[a^{<t-1>}, x^{<t>}] = \begin{bmatrix} a^{<t-1>} \\ \hline x^{<t>} \end{bmatrix}$$

100

10000

10100

$$[W_{aa} \,\vdots\, W_{ax}] \begin{bmatrix} a^{<t-1>} \\ x^{<t>} \end{bmatrix} = W_{aa}a^{<t-1>} + W_{ax}x^{<t>}$$

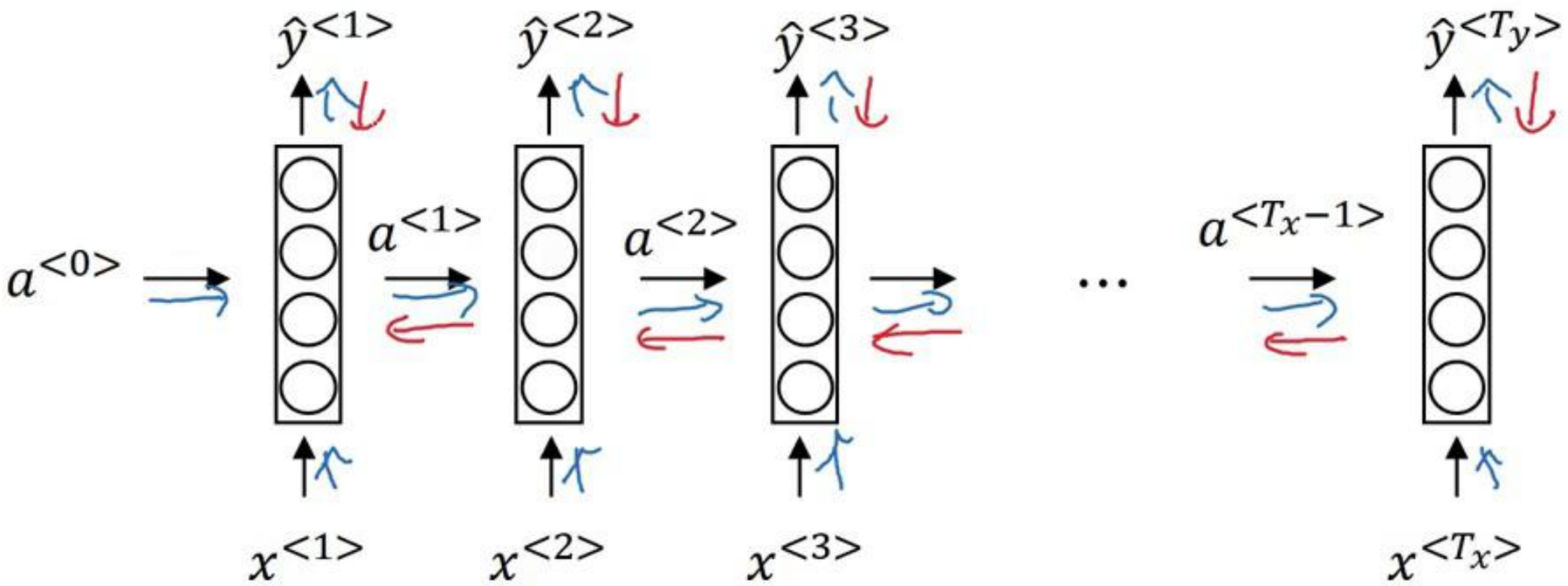
Andrew Ng

deeplearning.ai

Recurrent Neural Networks

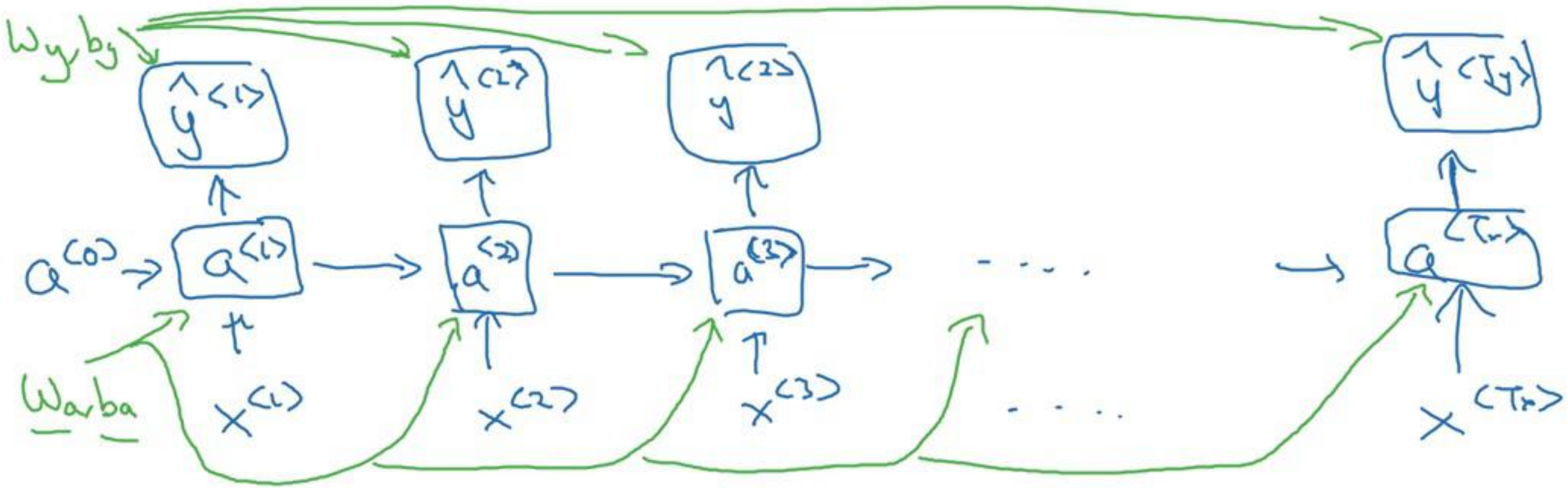
Backpropagation through time

Forward propagation and backpropagation



Andrew Ng

Forward propagation and backpropagation

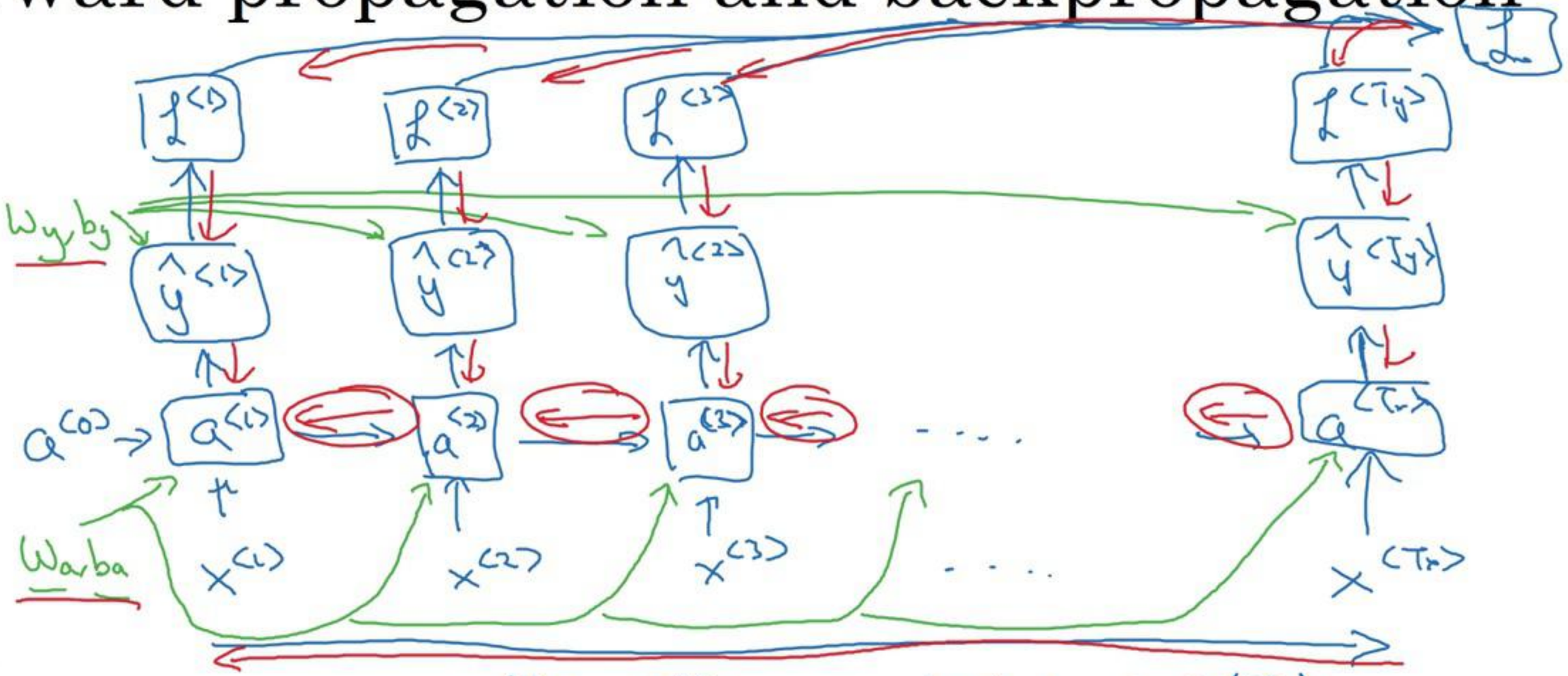


$$\mathcal{L}^{<t>}(\hat{y}^{<t>}, y^{<t>}) = -y^{<t>}\log\hat{y}^{<t>} - (1-y^{<t>})\log(1-\hat{y}^{<t>})$$

$$\mathcal{L} = \sum_{t=1}$$

Andrew Ng

Forward propagation and backpropagation



$$\mathcal{L}^{<t>}(\hat{y}^{<t>}, y^{<t>}) = -y^{<t>}\log\hat{y}^{<t>} - (1-y^{<t>})\log(1-\hat{y}^{<t>})$$

$$\mathcal{L}(\hat{y}, y) = \sum_{t=1}^{T_y} \mathcal{L}^{<t>}(\hat{y}^{<t>}, y^{<t>})$$

Backpropagation through time

Andrew Ng

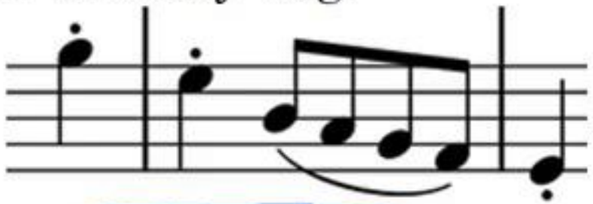
deeplearning.ai

Recurrent Neural Networks

Different types of RNNs

Examples of sequence data

T_x T_y

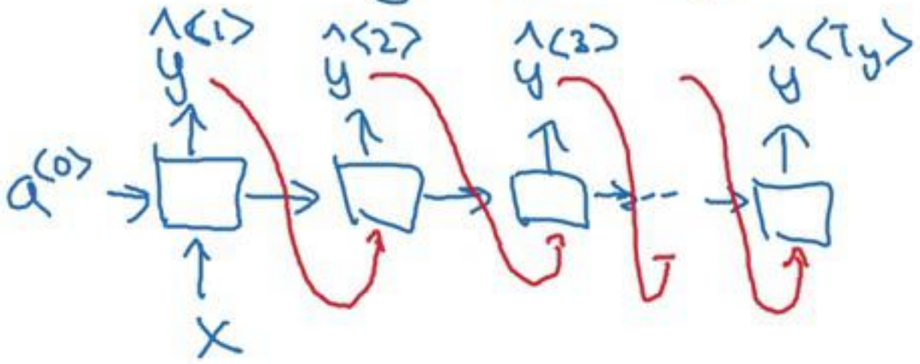
	x		y
Speech recognition		→	"The quick brown fox jumped over the lazy dog."
Music generation	∅	→	
Sentiment classification	"There is nothing to like in this movie."	→	★☆☆☆☆
DNA sequence analysis	AGCCCCTGTGAGGAACTAG	→	AGCCCCTGTGAGGAACTAG
Machine translation	Voulez-vous chanter avec moi?	→	Do you want to sing with me?
Video activity recognition		→	Running
Name entity recognition	Yesterday, Harry Potter met Hermione Granger.	→	Yesterday, Harry Potter met Hermione Granger.

Andrew Ng

Examples of RNN architectures

Music generation

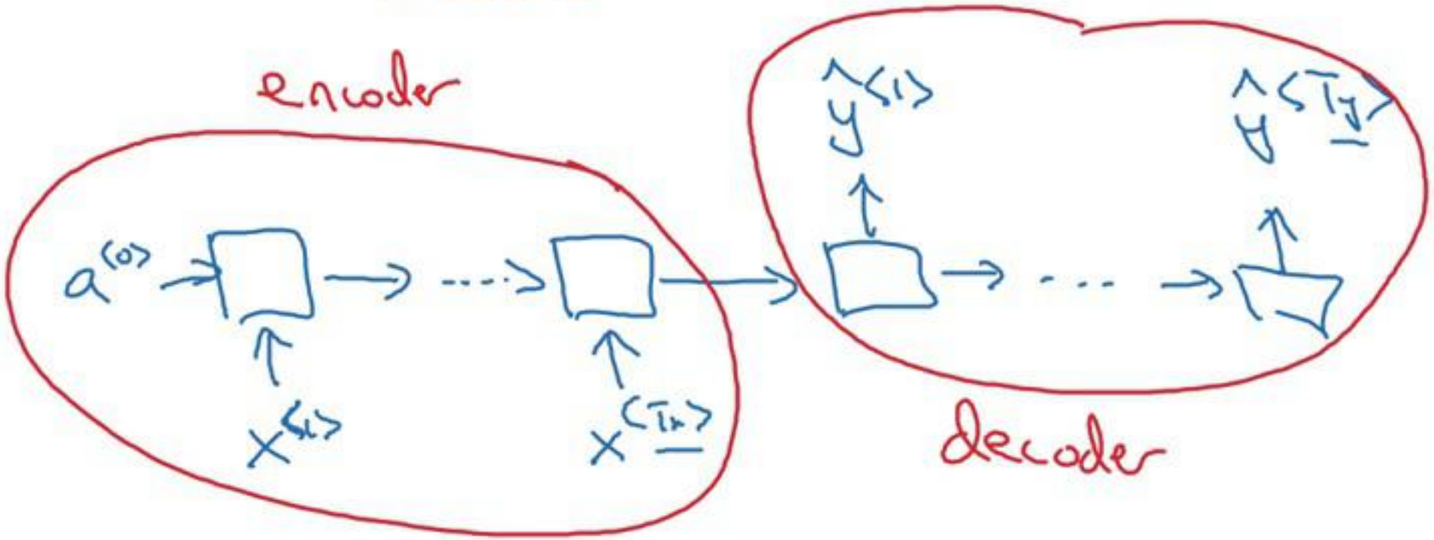
$x \rightarrow y^{<1>} y^{<2>} \ldots y^{<T_y>}$



One-to-many

$x = \phi$

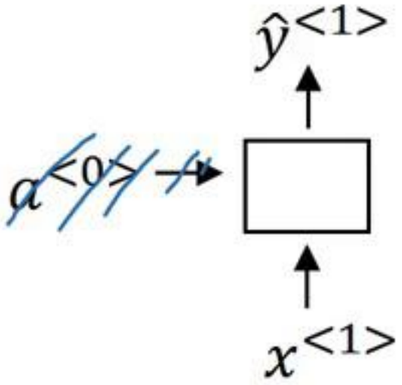
Machine translation



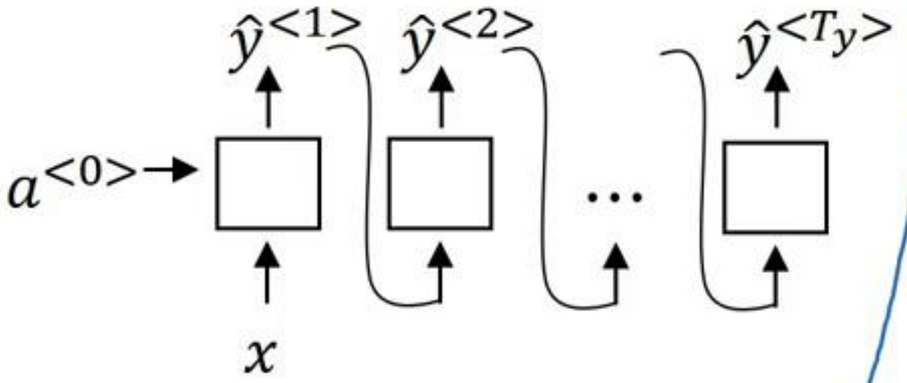
Many-to-many

Andrew Ng

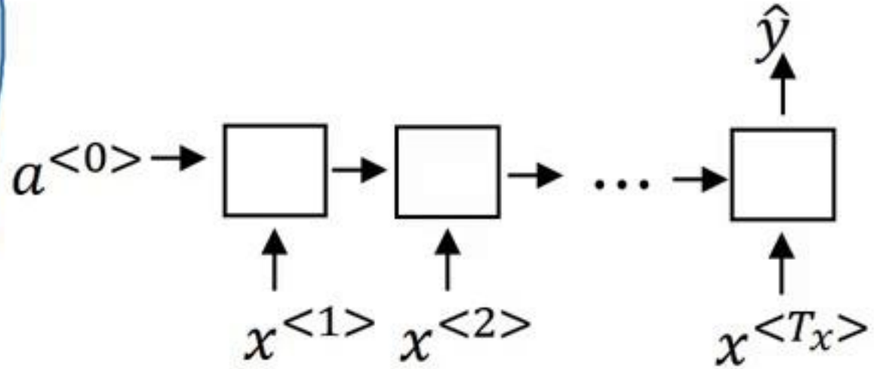
Summary of RNN types



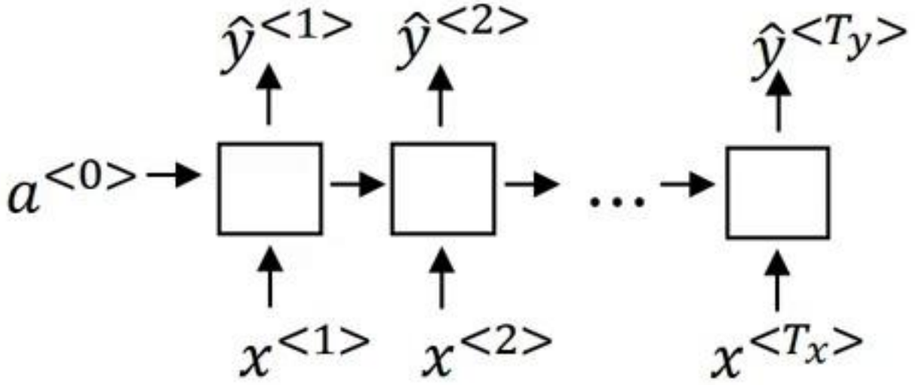
One to one



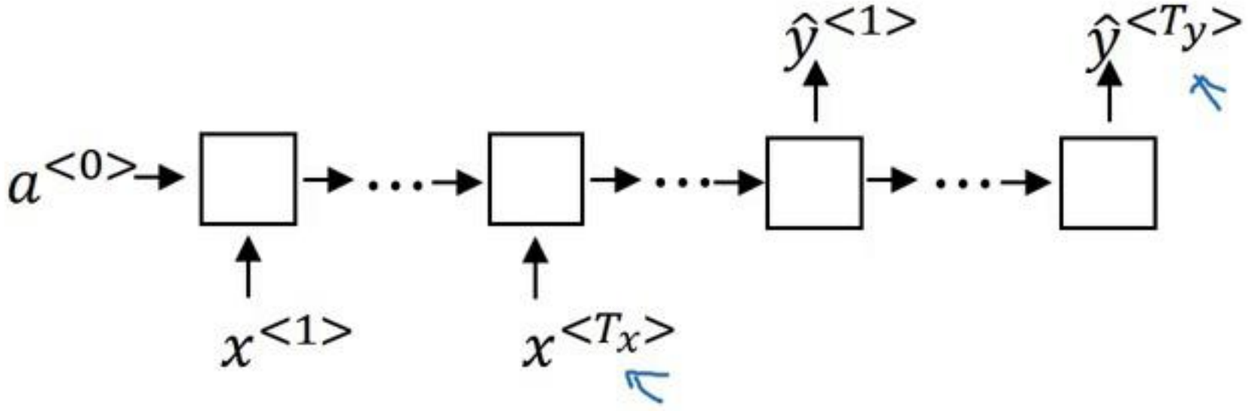
One to many



Many to one



Many to many $T_x = T_y$



Many to many

Andrew Ng

deeplearning.ai

Recurrent Neural Networks

Language model and sequence generation

What is language modelling?

Speech recognition

The apple and pair salad.

→ The apple and pear salad.

$P(\text{The apple and pair salad}) = 3.2 \times 10^{-13}$

$P(\text{The apple and pear salad}) = 5.7 \times 10^{-10}$

$P(\text{sentence}) = ?$

$P(y^{\langle 1 \rangle}, y^{\langle 2 \rangle}, \ldots, y^{\langle T_y \rangle})$

Andrew Ng

Language modelling with an RNN

Training set: large corpus of english text.

Tokenize

Cats average 15 hours of sleep a day. <EOS>

$y^{<1>}$ $y^{<2>}$ $y^{<3>}$ ... $y^{<8>}$ $y^{<9>}$

$x^{<t>} = y^{<t-1>}$

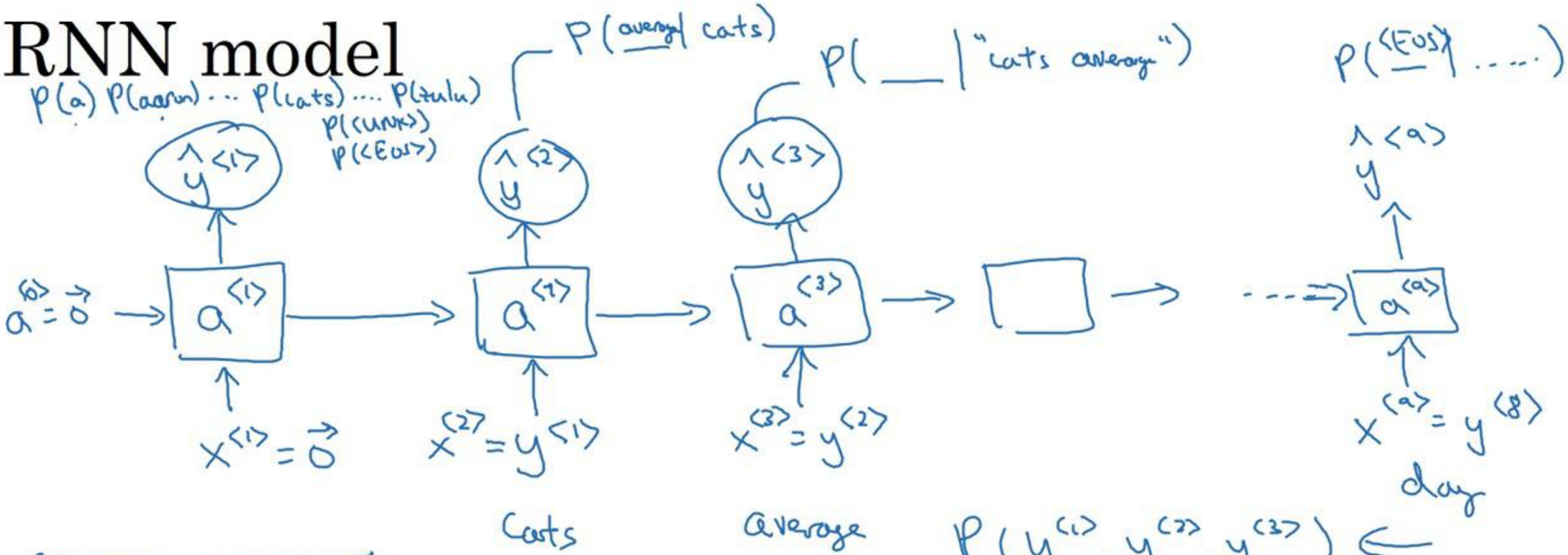
The Egyptian ~~Mau~~ is a bread of cat. <EOS>

<UNK>

10,000

Andrew Ng

RNN model



Cats average 15 hours of sleep a day. <EOS>

$$\mathcal{L}(\hat{y}^{<t>}, y^{<t>}) = -\sum_i y_i^{<t>} \log \hat{y}_i^{<t>}$$

$$\mathcal{L} = \sum_t \mathcal{L}^{<t>}(\hat{y}^{<t>}, y^{<t>})$$

$$P(y^{<1>}, y^{<2>}, y^{<3>})$$

$$= P(y^{<1>})\, P(y^{<2>} \mid y^{<1>})\, P(y^{<3>} \mid y^{<1>}, y^{<2>})$$

Andrew Ng

deeplearning.ai

Recurrent Neural Networks

Sampling novel sequences

Character-level language model

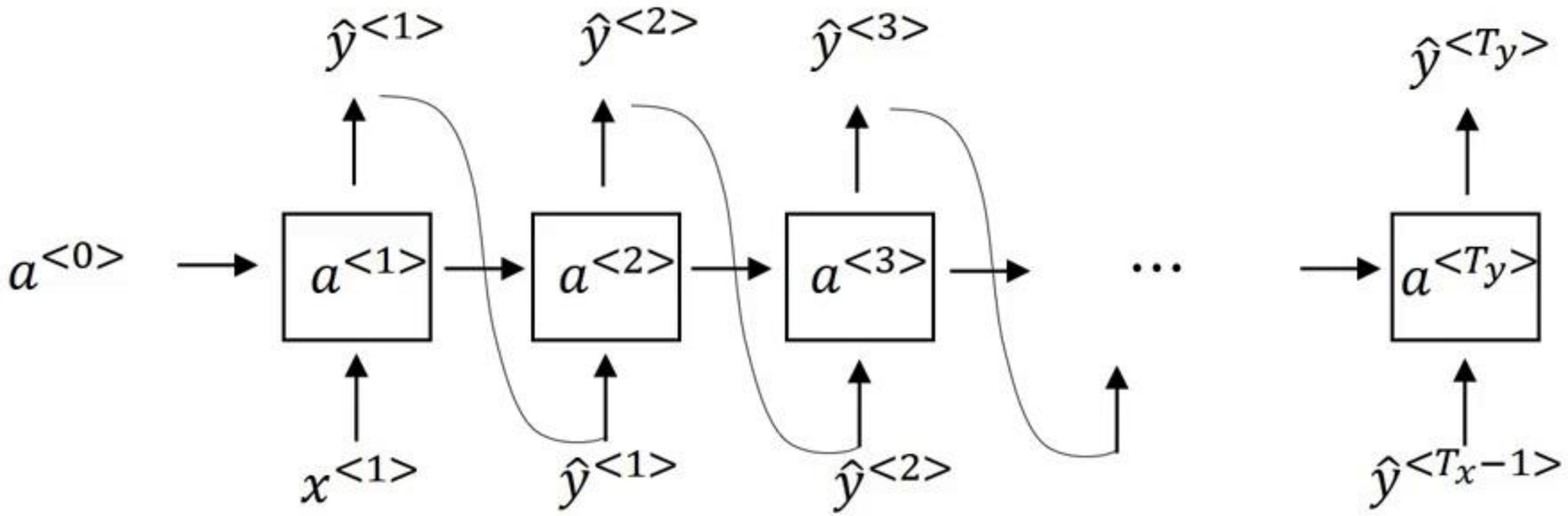
→ Vocabulary = [a, aaron, …, zulu, <UNK>] ←

→ Vocabulary = [a, b, c, ..., z, ␣, ., ,, ;, 0, ..., 9, A, ..., Z]

$y^{<1>}$ $y^{<2>}$ $y^{<3>}$ $y^{<4>}$

Cat average ...

Mau



Andrew Ng

Sequence generation

News

President enrique peña nieto, announced sench's sulk former coming football langston paring.

"I was not at all surprised," said hich langston.

"Concussion epidemic", to be examined.

The gray football the told some and this has on the uefa icon, should money as.

Shakespeare

The mortal moon hath her eclipse in love.

And subject of this thou art another this fold.

When besser be my love to me see sabl's.

For whose are ruse of mine eyes heaves.

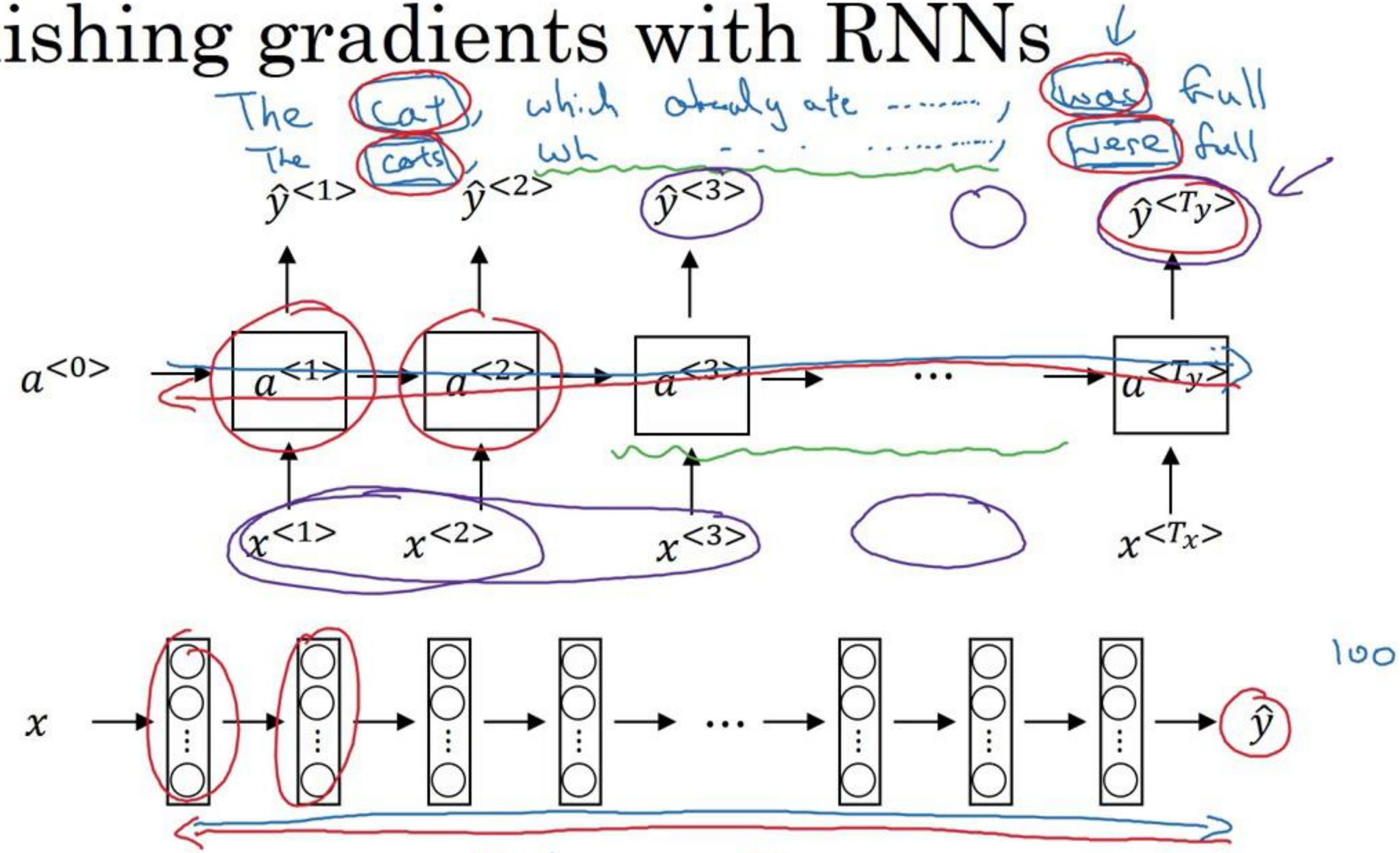
Andrew Ng

deeplearning.ai

Recurrent Neural Networks

Vanishing gradients with RNNs

Vanishing gradients with RNNs



Exploding gradients.

NaN

Gradient clipping

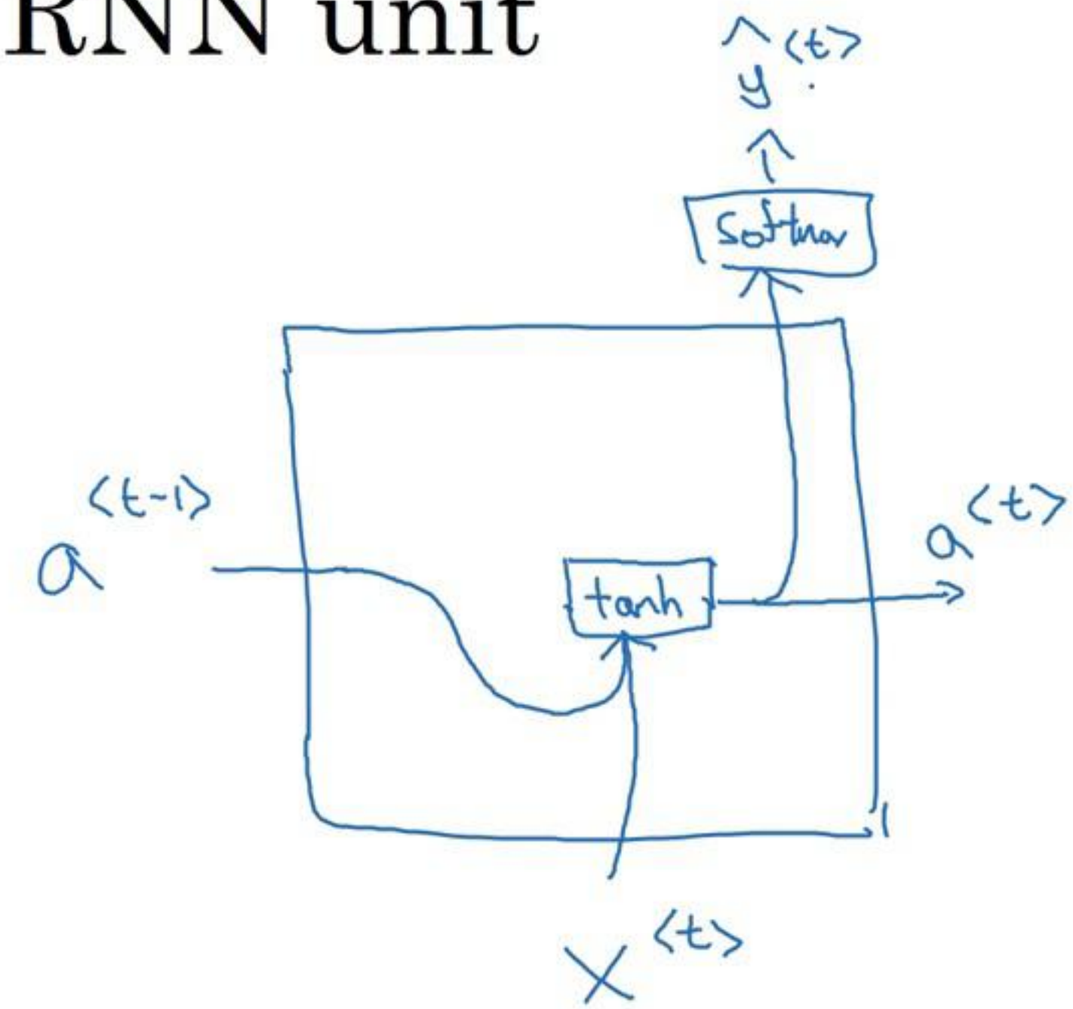
Andrew Ng

deeplearning.ai

Recurrent Neural Networks

Gated Recurrent Unit (GRU)

RNN unit

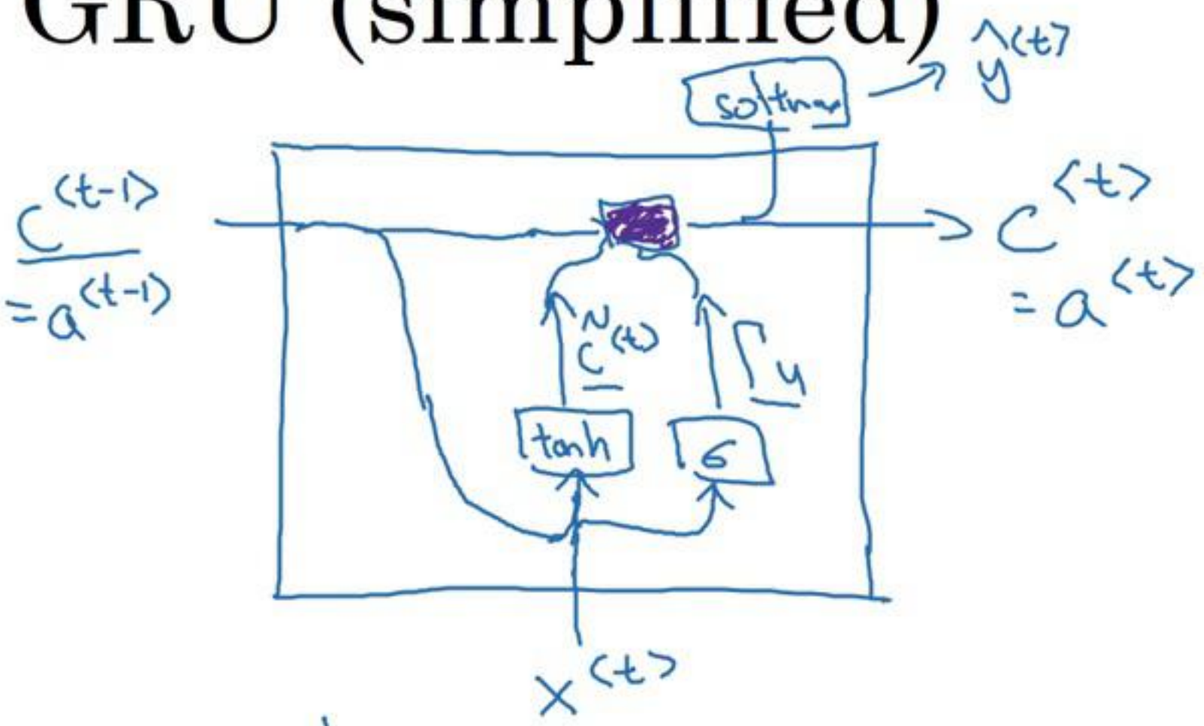


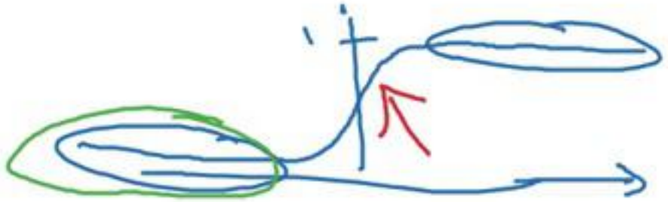
tanh

$$a^{<t>} = g(W_a[a^{<t-1>}, x^{<t>}] + b_a)$$

Andrew Ng

GRU (simplified)





c = memory cell

$$\boxed{c^{<t>}} = a^{<t>}$$

$$\boxed{\tilde{c}^{<t>}} = \tanh\left(W_c\left[c^{<t-1>}, x^{<t>}\right] + b_c\right)$$

$$\boxed{\Gamma_u} = \sigma\left(W_u\left[c^{<t-1>}, x^{<t>}\right] + b_u\right)$$

"update"

$$\boxed{c^{<t>}} = \Gamma_u * \tilde{c}^{<t>} + (1-\Gamma_u) * \boxed{c^{<t-1>}}$$

$\Gamma_u = 1$

element-wise

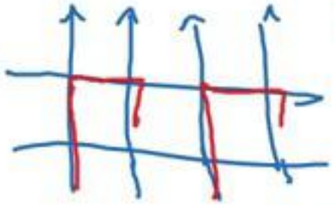
Gate

$\Gamma_u = 0.00001$

$\Gamma_u = 1$ $\Gamma_u = 0$ $\Gamma_u = 0$ $\Gamma_u = 0$ .....

$c^{<t>} = 1$ ..... $= 1$

The cat, which already ate …, was full.



[Cho et al., 2014. On the properties of neural machine translation: Encoder-decoder approaches]

[Chung et al., 2014. Empirical Evaluation of Gated Recurrent Neural Networks on Sequence Modeling]

Andrew Ng

Full GRU

$\tilde{h}$ $$\tilde{c}^{<t>} = \tanh(W_c[\Gamma_r * c^{<t-1>}, x^{<t>}] + b_c)$$

$$\begin{cases} u & \Gamma_u = \sigma(W_u[c^{<t-1>}, x^{<t>}] + b_u) \\ r & \Gamma_r = \sigma(W_r[c^{<t-1>}, x^{<t>}] + b_r) \end{cases}$$

LSTM

h $$c^{<t>} = \Gamma_u * \tilde{c}^{<t>} + (1 - \Gamma_u) + c^{<t-1>}$$

The cat, which ate already, was full.

Andrew Ng

Recurrent Neural Networks

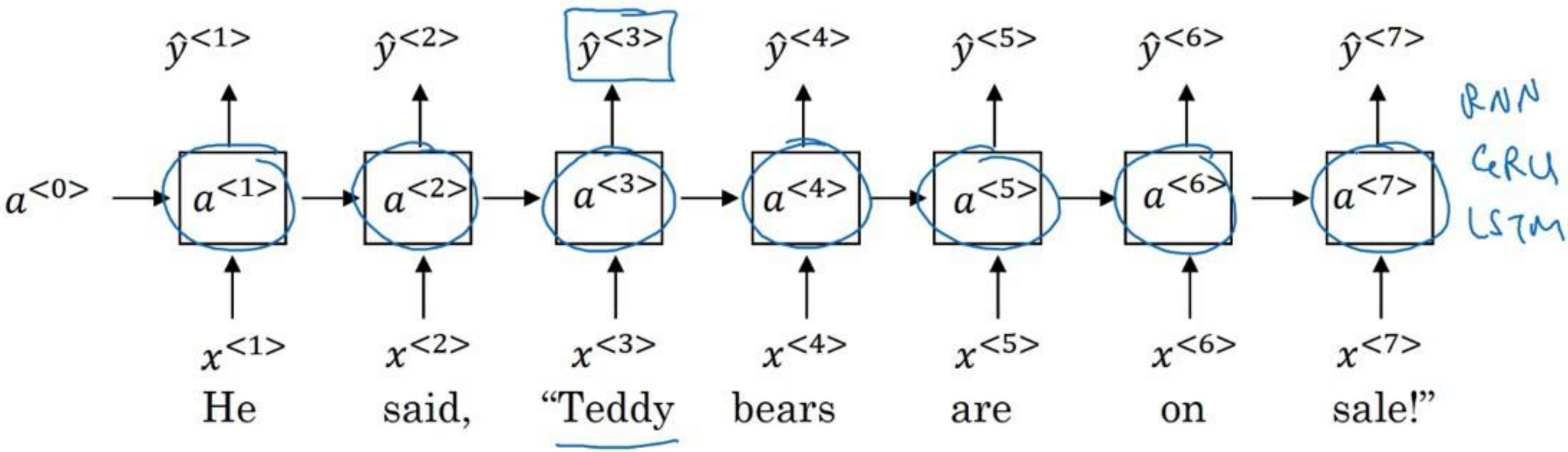
deeplearning.ai

Bidirectional RNN

Getting information from the future

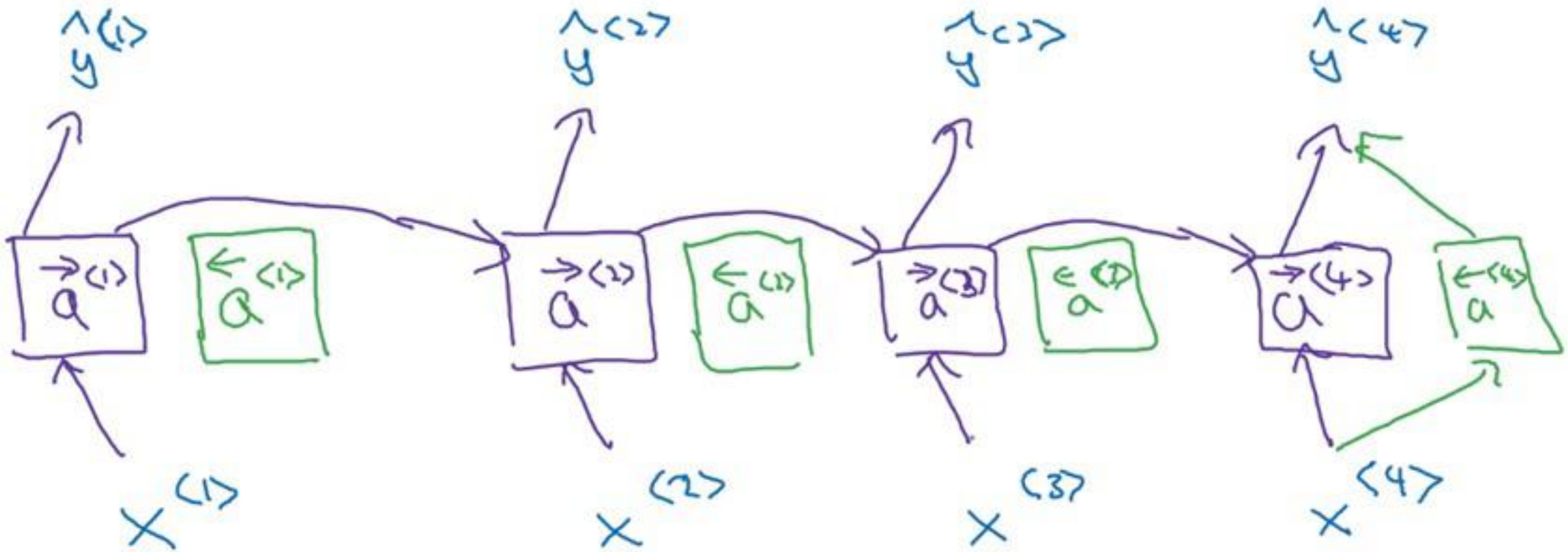
He said, "Teddy bears are on sale!"

He said, "Teddy Roosevelt was a great President!"



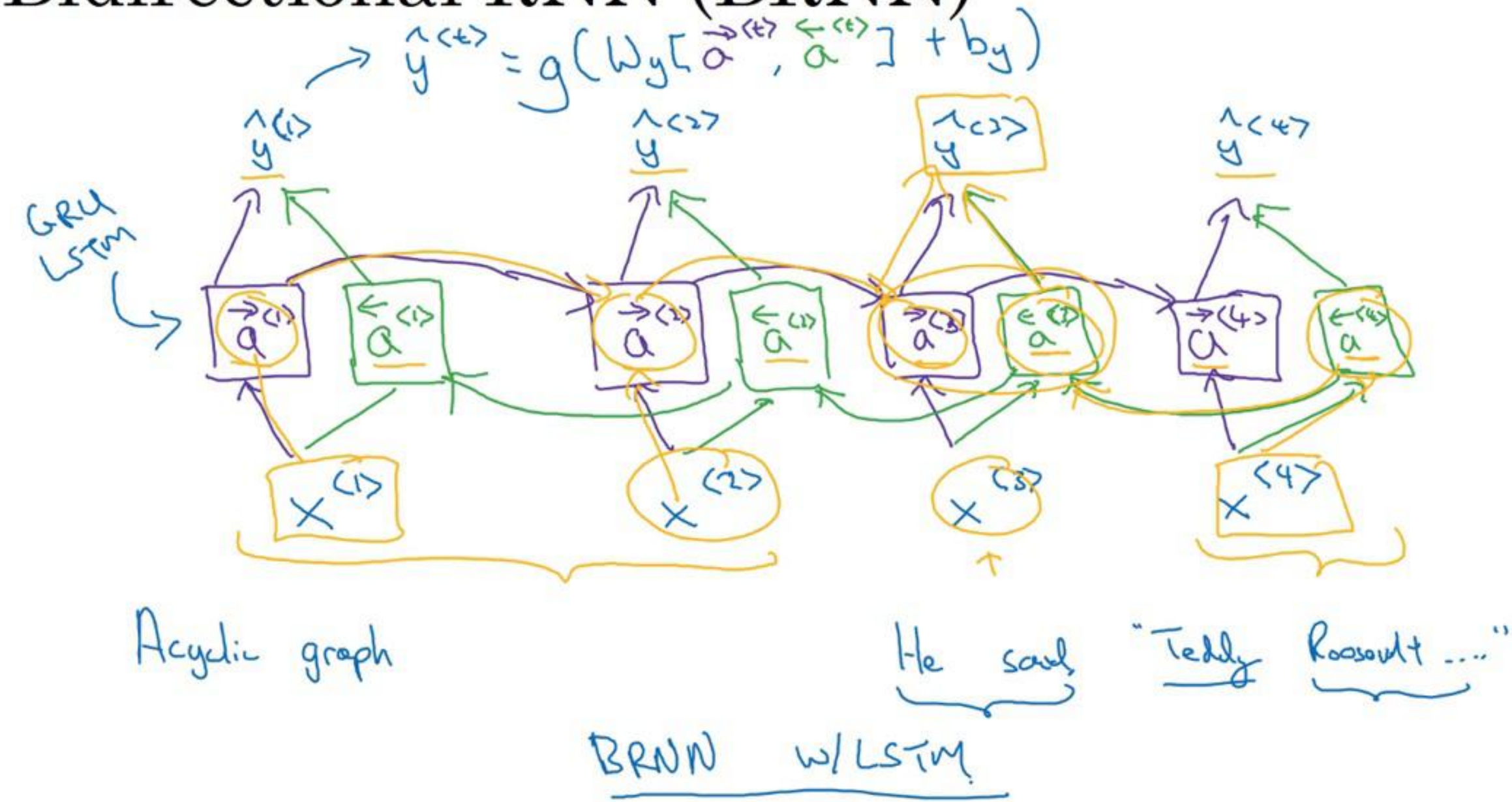
Andrew Ng

Bidirectional RNN (BRNN)



Andrew Ng

Bidirectional RNN (BRNN)



Andrew Ng

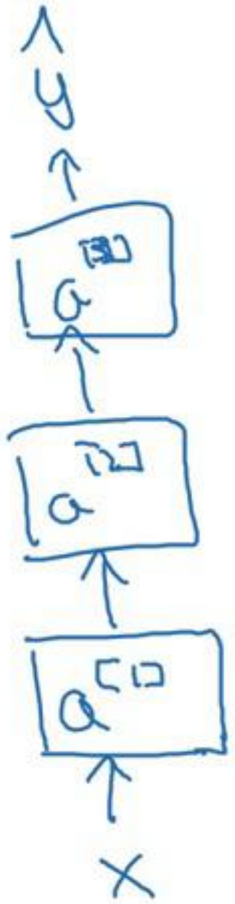
Recurrent Neural Networks

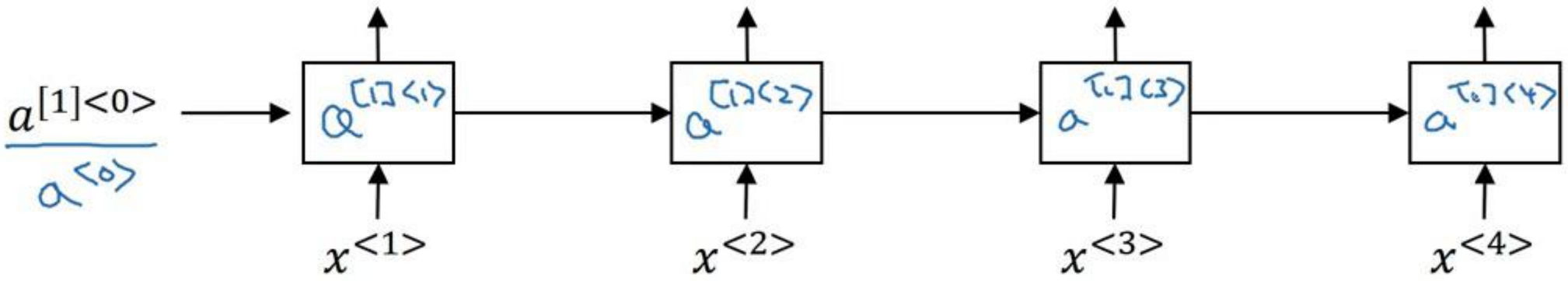
deeplearning.ai

Deep RNNs

Deep RNN example

$a^{[l]<t>}$

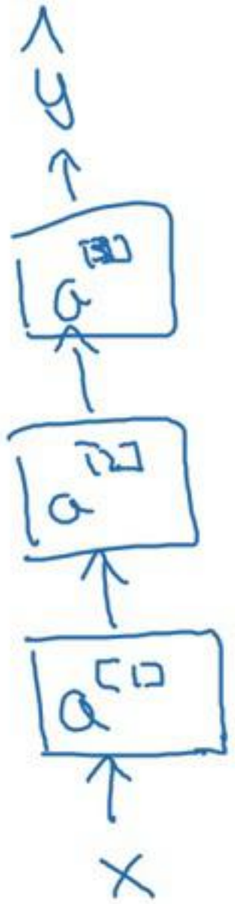


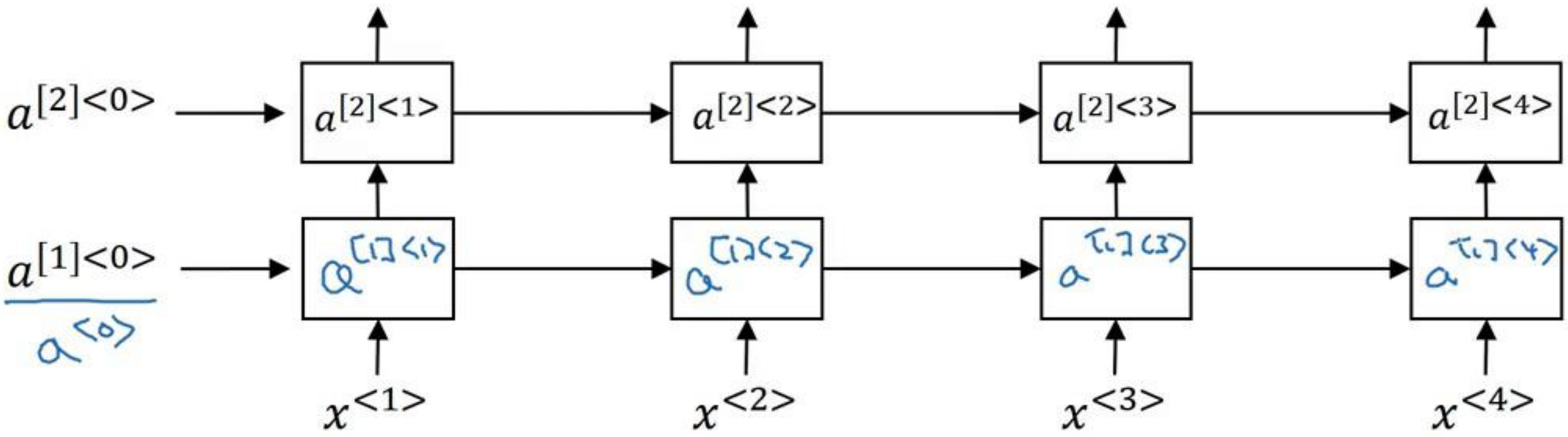


Andrew Ng

Deep RNN example

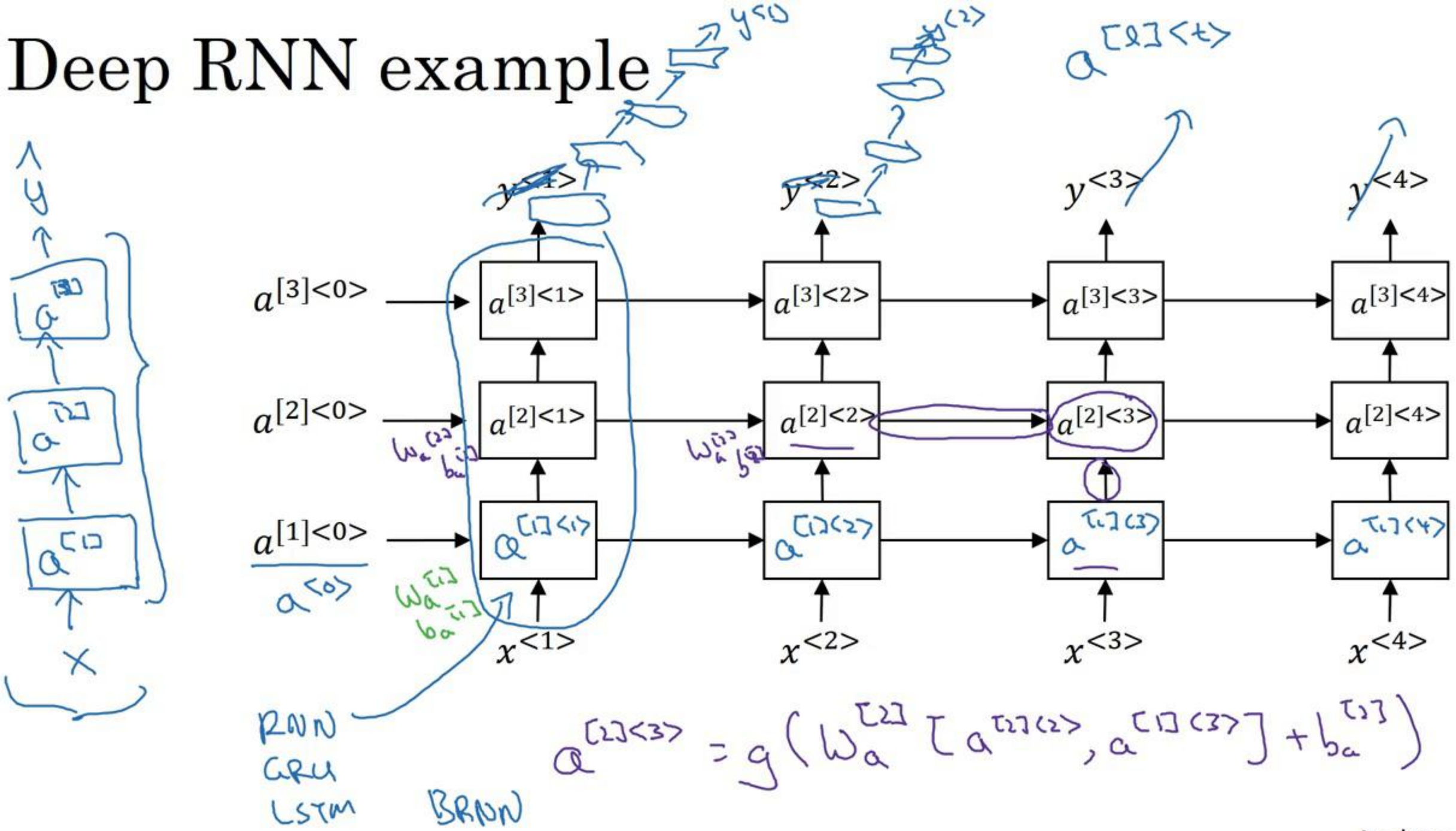
$a^{[l]<t>}$





Andrew Ng

Deep RNN example



$$a^{[2]<3>} = g\left(W_a^{[2]}\left[a^{[2]<2>}, a^{[1]<3>}\right] + b_a^{[2]}\right)$$

Andrew Ng